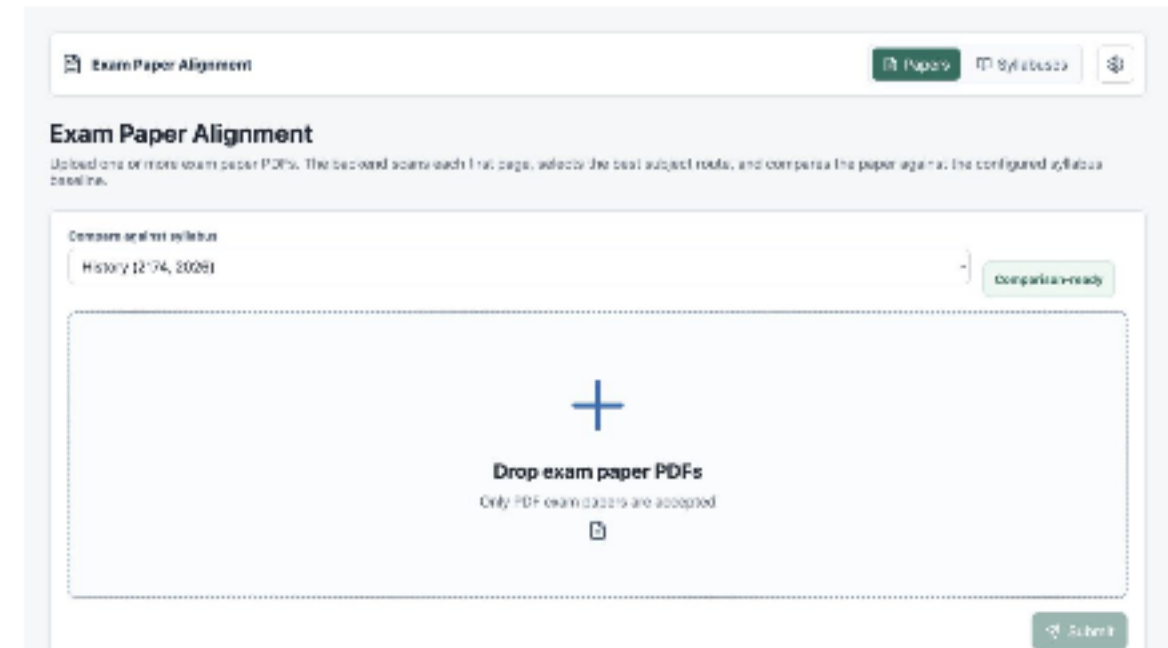


Exam Paper Analytical Tool

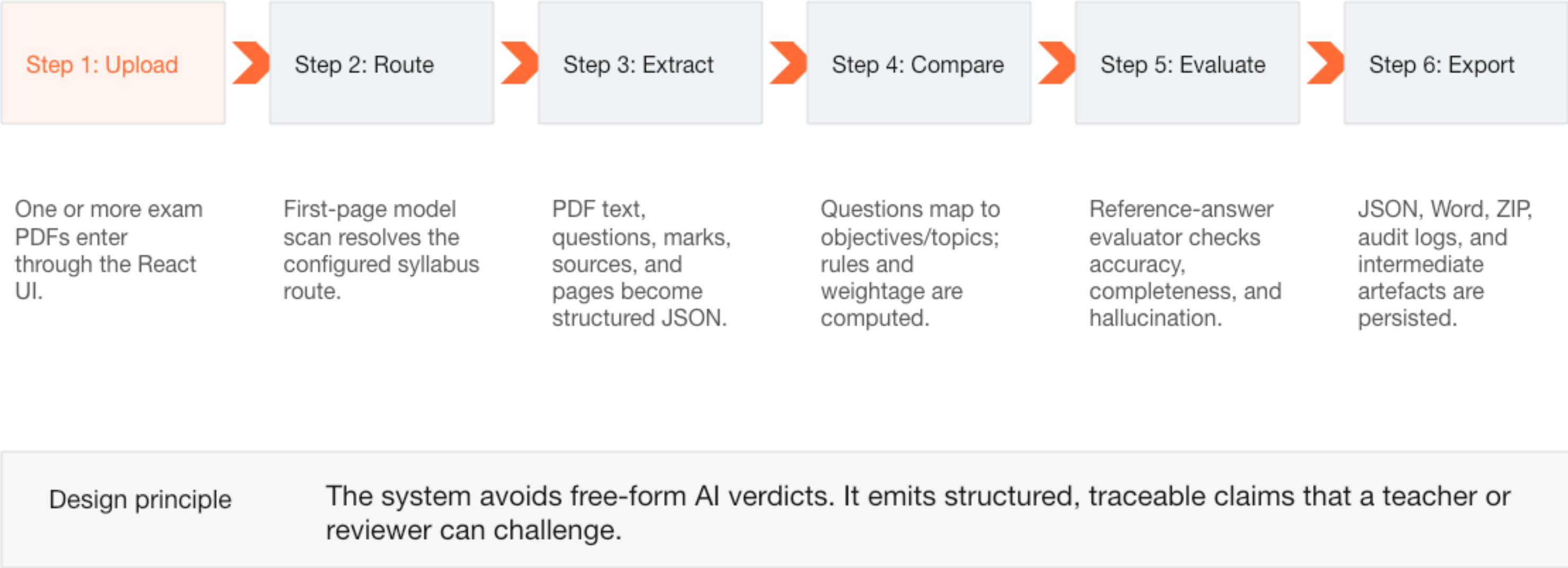
Syllabus-objective alignment, topic weightage balance, evaluation, and generalisation strategy

- React frontend for batch paper upload, syllabus management, LLM settings, and report download
- FastAPI backend for PDF ingestion, route resolution, extraction, alignment, weightage, and report artefacts
- Evaluation harness separates extraction success from AI mapping quality



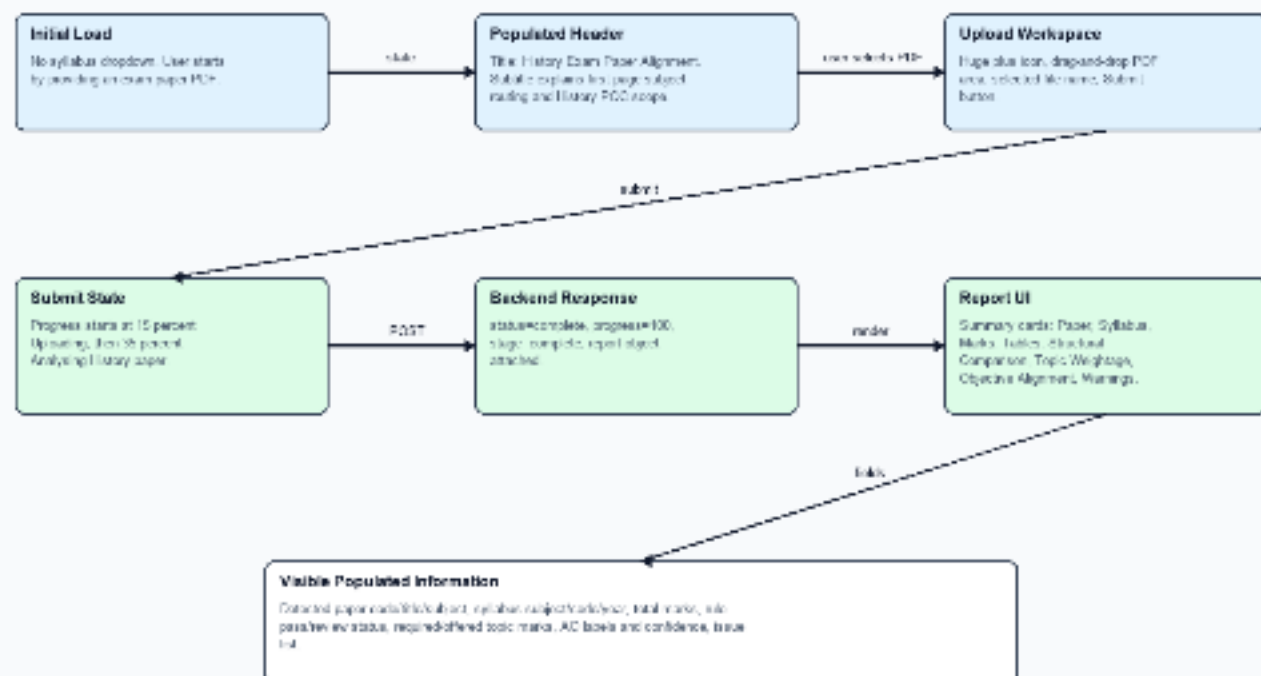
Management briefing with technical implementation detail

What Was Built



Frontend: Teacher-Facing Control Surface

Frontend Populated Flow

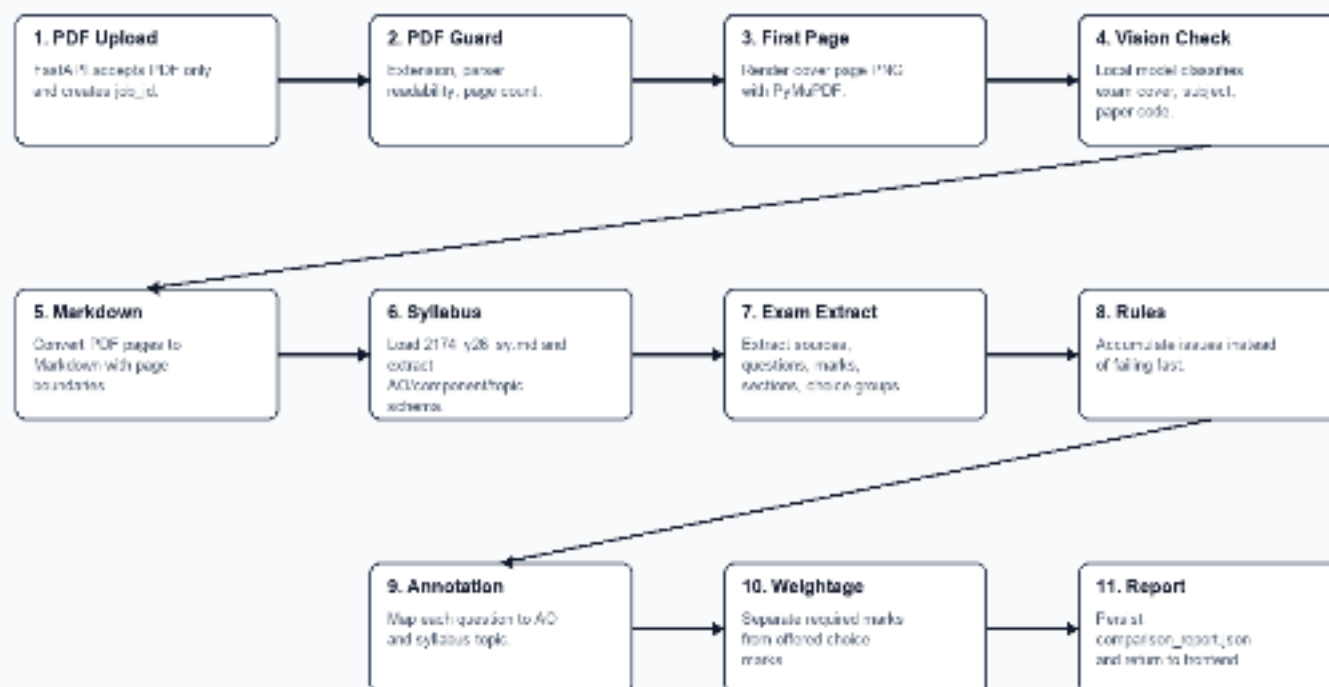


- Papers tab: PDF-only dropzone, batch upload, progress states, report tabs, JSON/Word/ZIP downloads.
- Syllabuses tab: SEAB URL or direct PDF ingestion with streamed event timeline and original PDF preview.
- Settings panel: model provider, base URL, model discovery, timeout, and connection test.
- Report view: structure, failed checks, topic weightage, objective alignment, and evidence pages.

Key files: App.tsx, client.ts, UploadDropzone.tsx, ComparisonReport.tsx, ModelSettingsPanel.tsx

Backend: Traceable Analysis Pipeline

Backend End-To-End Pipeline



Route resolution

First-page classification resolves configured syllabus route or stops with an explicit route issue.

Extraction and rules

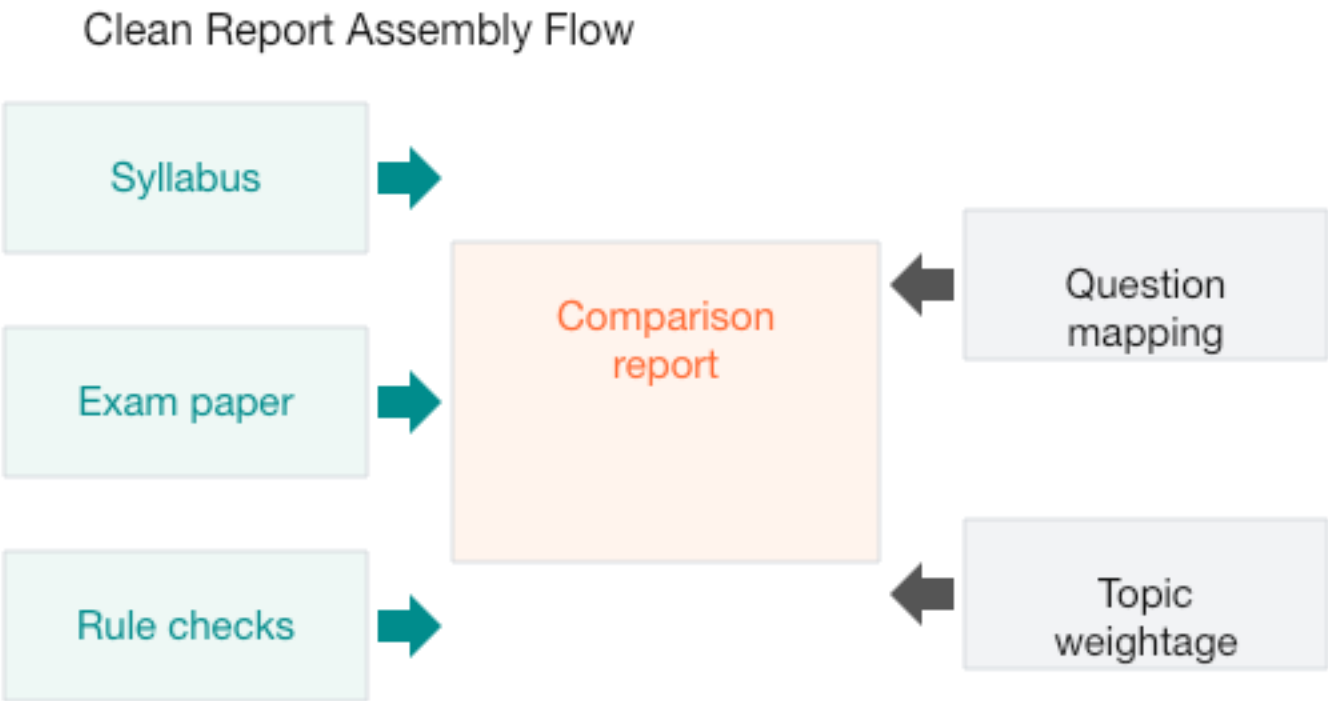
PDF Markdown plus syllabus JSON produce questions, sources, structure metrics, and rule checks.

Alignment and reports

Question annotations feed required/offered topic weightage and final ComparisonReport exports.

Data Contracts Make AI Claims Reviewable

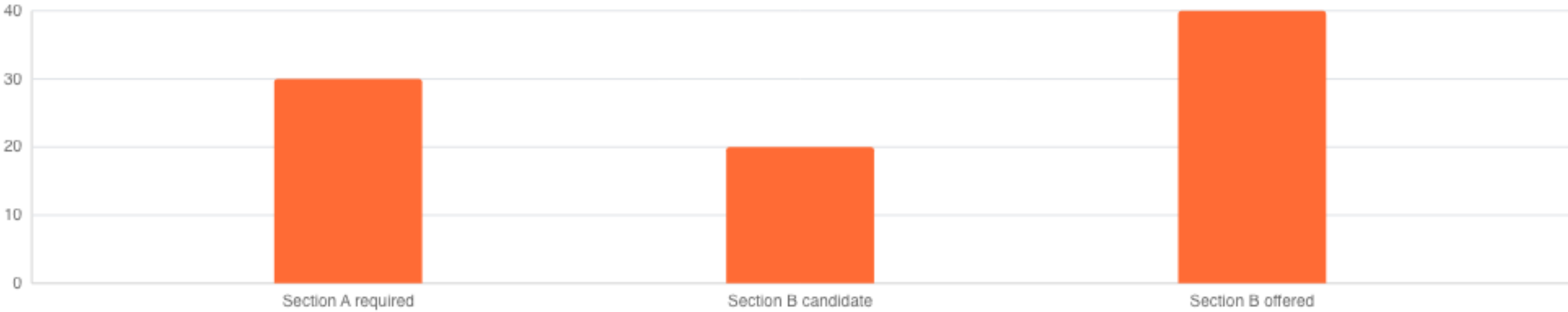
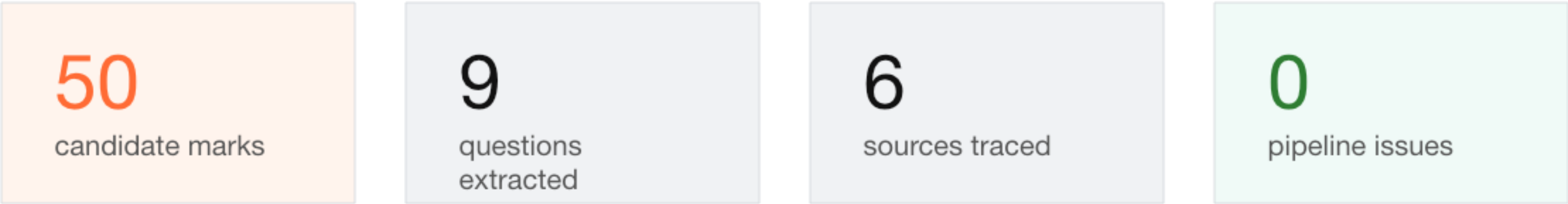
Object	Purpose	Review value
ExamPaper	Questions, sources, marks, pages	Grounds paper structure
SyllabusDocument	Objectives, components, topics	Defines baseline
Question Annotation	Objective/topic mapping plus evidence	Audits AI reasoning
TopicWeightage	Required vs offered marks	Shows topic balance
ComparisonReport	Checks, issues, metrics, downloads	Management-ready output



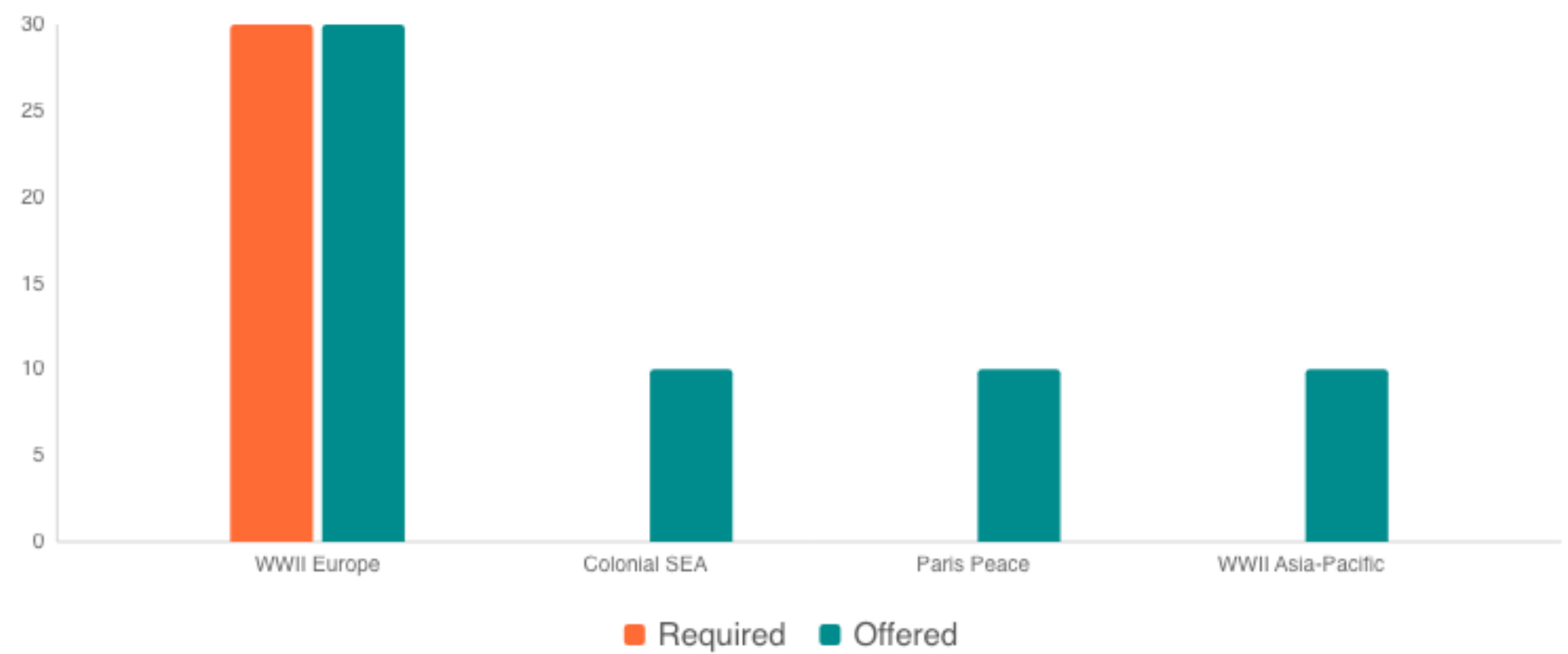
Inputs and checks feed one final report payload.

Persisted outputs include original upload, first-page image, Markdown, raw model response, raw extraction, exam structure, report JSON, and audit log.

History Paper 1 Result: Structure Extracted Cleanly



Topic Weightage: Required vs Offered Marks



Interpretation

Section A concentrates compulsory marks on WWII Europe.

Reviewer action

Confirm whether this is intended for the paper or should be diversified in future papers.

Note: reviewed reference mappings illustrate the expected AI output; fresh mock runs correctly expose mapping uncertainty.

Evaluation: What The Scorer Catches

Run	Exact	Complete	Hallucination	Meaning
Reference set	1.0000	1.0000	0.0000	Target output matched
Fresh mock	0.2143	0.5357	0.3214	Extraction passes; mapping uncertain

- Accuracy: exact and textual similarity against expected structured keys.
- Completeness: required fields are present, including objectives and topic-weightage rows.
- Hallucination: predicted values must be supported by paper text or the reviewed reference answer.
- Operational lesson: model output should be evaluated separately from pipeline health.

Generalising Beyond History

Reusable

- PDF validation
- First-page routing
- Markdown extraction
- Syllabus ingestion
- LLM JSON prompts
- Report exports

Subject-specific

- Assessment objectives
- Component rules
- Choice policies
- Topic taxonomy
- Evidence expectations
- Reference answer sets

Production controls

- Self-hosted model
- Prompt versioning
- Teacher review queue
- Audit logs
- Data retention
- Fallback behaviour

Conclusion: the architecture generalises, but each subject needs a reviewed rule pack and answer set before high-stakes use.

Practical Usage And Next Steps

- Use as a teacher-review assistant, not an autonomous approval engine.
- Run self-hosted models through LM Studio, Ollama, or OpenAI-compatible endpoints for local governance.
- Require evidence pages and ambiguity notes for all alignment claims.
- Add subject packs incrementally: syllabus JSON, extraction profile, rule checks, prompt tests, reference answer set.
- Improve next: real-model benchmark set, UI screenshots, reviewer feedback loop, and confidence calibration.



Recommended governance

Keep human sign-off, preserve audit logs, and review uncertain mappings before acting.